Module 1 — Read 1

Probability language through the binomial pmf

Module 1, Read 1: Probability for a binomial test

Project: Project 1 — Testing a Claim · **Preview:** Read 0 (sample tour) · **Next on the course path:** Quiz A (Def & Notation), then Read 2.

Optional supplements :

- Textbook by **Vu & Harrington**, see Ch 2 — Probability and Ch 3.2 — Binomial;
- Videos: Khan: binomial distribution · Khan: visualizing binomial.
- ChatGPT or Notebook LM: give this page to a genAI and ask it to step you through concepts and test you with the questions to prepare for the quiz. Ask it to cover each of the learning outcomes.

If you have not skimmed the sample ESP report yet, start with **Read 0** (~5 minutes). This reading is your **self-paced textbook** for Module 1, Part 1. Work one **section** at a time; answer the **Check yourself** boxes before moving on (same ideas appear on Quiz A).

Learning outcomes

Work through all seven sections below in order. Each section has **Check yourself** boxes before you move on.

After this reading you should be able to:

Section 1: Data, notation, and probability

Probability as data vs theory

- Compute counts, proportions, and percentages for an event across repeated trials.
- Use symbols x, n, $\hat{p}$, and $P(\text{event})$ for data vs theoretical probability.
- Distinguish observed outcomes from theoretical probability in a random process.
- State three rules that probabilities must satisfy.
- Define probability and connect it to long-run relative frequency.

Section 2: Sample spaces and equally likely outcomes

Probability calculations the long way — sample spaces and events

- Define and identify **sample space**, **outcome**, and **event** for a trial.
- Use notation $P(A)$ for the probability of an event.
- State and use the formula for $P(A)$ when outcomes in the sample space are equally likely.

Section 3: Addition rule

Addition rule for probability

- State the **addition rule** for any two events A and B.
- Use the addition rule to compute $P(A \text{ or } B)$, including overlapping events.
- Apply special cases: **complement rule** and **non-overlapping** events.

Section 4: Conditional probability and independence

Conditional probability and multiplication rule

- State the definition of **conditional probability** and compute $P(A \mid B)$.
- Use the **multiplication rule** to compute $P(A \text{ and } B)$.

- Decide whether two events are **independent** using equivalent criteria.

Section 5: Counting and combinations

Permutations and combinations

- Distinguish settings where **order matters** (permutations) from where it does not (combinations).
- Use ${}_nP_r$ and ${}_nC_r$ (or $\binom{n}{r}$) to solve counting problems.
- Relate factorials to the permutation and combination formulas.

Section 6: Random variables and the binomial process

From coin tosses to the binomial random process

- Define a **discrete random variable** and its **probability distribution (pdf)**.
- Define a **binomial random variable** and a **binomial random process**.
- Match notation (n, x, p) and the **four conditions** for a binomial process to a coin-toss setting.
- Express $P(X = x)$ using the binomial pdf.

Section 7: Binomial pmf

Practice — binomial random process

- Decide whether a scenario satisfies the **four binomial conditions** and justify your answer.
- Identify n, p, and the random variable X for a binomial count.
- Write and evaluate probabilities using the binomial pdf; use mean $\mu = np$ and SD $\sigma = \sqrt{np(1-p)}$ as needed.

Section 1: Data, notation, and probability

Learning Outcomes:

By the end of today's lecture, you should be able to:

1. Compute counts, proportions, and percentages of a particular event occurring in a number of trials.
2. Use symbols ($x, n, \hat{p}, P(\text{event})$) correctly to describe observed data and theoretical probability.
3. Distinguish between observed outcomes (data) and theoretical probability.
4. Describe three rules that probabilities must satisfy.
5. State the definition of probability and connect it to the idea of long-run relative frequency in a repeatable random process.

Physical Randomness: Tossing a Coin

> ℹ Example in-class data (illustrative)
>
> Suppose three students record: Rosanna 7 heads in 10 tosses ($\hat{p} = 0.7$); You 4 heads in 10 ($\hat{p} = 0.4$); Neighbor 6 heads in 12 ($\hat{p} = 0.5$). Counts and proportions differ because n and randomness differ.

Let's toss a coin a few times in class, keeping track of the tosses that land "heads."

Rosanna's data

- Count of heads (x): **7**
- Total tosses (n): **10**
- Proportion ($\hat{p}$): **0.7**

- Percentage: **70%**

Your data

- Count of heads (x): **4**
- Total tosses (n): **10**
- Proportion ($\hat{p}$): **0.4**
- Percentage: **40%**

Your neighbor's data

- Count of heads (x): **6**
- Total tosses (n): **12**
- Proportion ($\hat{p}$): **0.5**
- Percentage: **50%**

What's the highest count of heads? **7** (Rosanna)

Most tosses? **12** (neighbor)

Highest proportion of heads? **0.7** (Rosanna)

Highest percentage of heads? **70%** (Rosanna)

Comments:

If we care about comparing coins (with the same number of tosses), then $\hat{p}$ **(proportion)** is the best summary.

If we care about getting lots of "heads" then x **(count)** is the best summary.

The summary of total tosses (n) tells us **how much data we collected / how precise the proportion can be**.

Question: Rosanna keeps tossing her coin forever. What would we expect?

- Data (what happened): **varies** from sample to sample (e.g. $\hat{p}$ bounces around)
- Theory (long run): **approaches 0.5** for a fair coin

Key definitions related to a trial

Fill in each definition in your own words:

- **Sample space:**

- **Outcome:**

- **Event:**

❗ Notation: Data vs theory

Symbols and notation

We'll use two colors: one for observed data and another for theoretical quantities.
Async note: In class we color-code **theoretical** vs **observed** quantities. On your own, pick two highlighters and use them consistently for $P(\text{event})$ vs $\hat{p}$.

- x = count of "successes" in n observed trials
- n = number of observed trials
- $\hat{p} = x/n$ = observed proportion of successes in our trials
- Percentage of observed trials that were a success $= 100 \cdot \hat{p}$
- $P(\text{event})$ = probability of an event

The notation for probability, $P(\text{event})$, is in the same format as notation for functions in high school algebra:

- $=$ reads as **"is"** (or "equals")
- $()$ reads as **"of"** — the event inside the parentheses (not multiplication!)

$Y = f(x)$ reads as "y is the value of the function whose name is f when x is put in."
$Y = P(A)$ reads as "y is the value of probability when the event under consideration is A."

Example 1: Coin tosses

Suppose we toss a coin 20 times and see 15 heads.

- $x = 15$
- $n = 20$
- $\hat{p} = 15/20 = 0.75$
- Percentage = 75%
- Theoretical $P(\text{Heads}) = 0.5$ (for a fair coin)

Student exercise 1: Coin tosses

A coin is tossed 10 times and comes up heads 3 times. Fill in:

- $x = 3$
- $n = 10$
- $\hat{p} = 3/10 = 0.3$
- Percentage = 30%
- Theoretical $P(\text{Heads}) = 0.5$

Example 2: Die rolls

Suppose a fair 6-sided die is rolled 24 times and we see the number "5" exactly 2 times.

- $x = 2$
- $n = 24$
- $\hat{p} = 2/24 \approx 0.083$
- Percentage $\approx 8.3\%$
- Theoretical $P(\text{"5"}) = 1/6$

Student exercise 2: Die rolls

A die is rolled 30 times and shows "1" exactly 10 times. Fill in:

- $x = 10$
- $n = 30$
- $\hat{p} = 10/30 \approx 0.333$
- Percentage $\approx 33.3\%$
- Theoretical $P(\text{"1"}) = 1/6$

! Rules for probabilities (axioms)

All probabilities must satisfy:

1. $0 \leq P(A) \leq 1$ for any event A
2. Total probability across all possible outcomes $= 1$
3. If A and B are two events with no overlap, then $P(A \text{ or } B) = P(A) + P(B)$

Examples

- Coin: $P(\text{Heads}) + P(\text{Tails}) = \mathbf{1}$
- Die: $P(\text{"1"}) + P(\text{"2"}) + \cdots + P(\text{"6"}) = \mathbf{1}$
- Die: $P(\text{"1"}) + P(\text{"2"}) = 1/3$
- $P(A) = 70$ is **invalid** (probabilities must be between 0 and 1)
- $-0.5 = P(B)$ is **invalid** (cannot be negative)
- $P(D) = 0.7$ is **valid**

! Definition: Probability

$P(\text{Event})$ is the proportion of times the event occurs out of the number of trials as the trial is repeated infinitely many times.
Probability is tied to:

- A specific random process (repeating a trial with uncertain outcome infinitely many times)

- Long-run relative frequency (a.k.a. "proportion") of an event occurring

Check yourself — Data, notation, and probability

1. A fair coin is tossed 20 times and shows 9 heads. What are x, n, and $\hat{p}$?
2. In one or two sentences, what is the difference between $\hat{p}$ and $P(\text{event})$?
3. What is the definition of $P(\text{Event})$ in terms of long-run relative frequency?
4. State the three rules (axioms) that all probabilities must satisfy.
5. Define **sample space**, **outcome**, and **event**.
6. A coin is tossed 10 times with 3 heads. Compute $\hat{p}$ and the percentage of heads.

Answers — Data, notation, and probability

1. $x = 9$, $n = 20$, $\hat{p} = 9/20 = 0.45$.
2. $\hat{p}$ is the **observed proportion from data** in your sample. $P(\text{event})$ is the **long-run theoretical probability** for the random process.
3. $P(\text{Event})$ is the proportion of times the event occurs when the trial is repeated infinitely many times.
4. (1) $0 \leq P(A) \leq 1$. (2) $P(S) = 1$. (3) If A and B do not overlap, $P(A \text{ or } B) = P(A) + P(B)$.
5. **Sample space** S: all possible outcomes of a trial. **Outcome**: one element of S. **Event**: a set of one or more outcomes (often written A, B, …).
6. $\hat{p} = 3/10 = 0.3$; percentage $= 30\%$.

Section 2: Sample spaces and equally likely outcomes

Probability calculations the "long way"

Learning Objectives

- Define and identify the following terms for a given trial:
  - Sample space
  - Outcome
  - Event
- Use appropriate notation for the probability of an event, $P(A)$
- State and use the formula for the probability of an event if the outcomes in the sample space are equally likely.

Definitions

Let's define the following terms:

> ! Definition: Sample space
>
> The **sample space** S is the set of all possible outcomes of a trial.
>
> ## Definition: Outcome
>
> An **outcome** is one element of the sample space (one possible result of a trial).
>
> ## Definition: Event
>
> An **event** is a set of one or more outcomes of a trial, usually written with an uppercase letter such as A, B, or H (for "heads").

Set Notation and Operator Practice

Fill in the meaning or notation for each of the following:

Suppose that a single six-sided die is rolled.

Worked solution — set notation (one die roll)

- $S = \{1, 2, 3, 4, 5, 6\}$
- Let $A = \{5, 6\}$ (die shows 5 or 6)
- Not A means A^c — the die shows 1, 2, 3, or 4
- $B = \{2, 4, 6\}$ (even)
- A and $B = \{6\}$
- A or $B = \{2, 4, 5, 6\}$
- A given B: assuming B occurred, what is the chance A also occurred?

Probability Notation

Translate the following "sentences" to English:

- $P(A) = 0.7$
- $P(A \text{ and } B) = 0.2$
- $P(\text{not } A) = 0.1$
- $P(A|B) = 0.25$

Worked translations — probability notation

- $P(A) = 0.7$: 70% chance event A occurs.
- $P(A \text{ and } B) = 0.2$: 20% chance **both** A and B occur.
- $P(\text{not } A) = 0.1$: 10% chance A does **not** occur.
- $P(A|B) = 0.25$: given B occurred, there is a 25% chance A also occurs.

Three "Axioms" of Probability

Fill in the blanks:

Worked solution — probability axioms

1. $0 \leq P(A) \leq 1$
2. $P(S) = 1$
3. $P(A \text{ or } B) = P(A) + P(B)$ when A and B do not overlap

> **Note:** "axiom" is a type of rule that we must merely "agree" is true, rather than being able to "prove" that it is true.

Translate the following sentence to correct probability notation:

"The chance of it raining in the next 10 minutes is 80%"

Answer: $P(\text{rain in next 10 minutes}) = 0.80$

Rule: Equally likely outcomes

If all outcomes in the sample space S of a trial are equally likely, then

$$P(A) = \frac{|A|}{|S|}$$

where A is any event that is contained in S, and the notation |set| is the **number** of outcomes in the set.

Counting strategy (Sections 2–4)

Until Section 5 introduces factorials, permutations, and combinations, find probabilities by **listing outcomes in** S (or listing in a clear pattern) and **counting** how many belong to the event. Then use $P(A) = |A|/|S|$. That habit is what Section 5 formulas will speed up.

Coin Toss Example

Suppose a fair coin is tossed 2 times.

1. List the sample space S of this experiment
2. List the outcomes in the sample space that are in the event B = "the coin lands heads at least once"
3. Find $P(B)$.

Worked solution — two coin tosses

1. $S = \{\text{HH, HT, TH, TT}\}$
2. $B = \{\text{HH, HT, TH}\}$ ("heads at least once")
3. $P(B) = 3/4$

Six-sided Fair Die Example

Suppose a fair die is rolled twice.

1. List the sample space S of this experiment
2. List the outcomes in the sample space that are in the event C = "the sum of the two rolls is 7"
3. Find $P(C)$

Worked solution — two dice (sum is 7)

1. List S in a table: **rows = die 1**, **columns = die 2**. Each cell is one ordered pair $(\text{die }1, \text{die }2)$ — **36** equally likely outcomes.

Die 1	Die 2	1	2	3	4	5	6
1		(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2		(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3		(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4		(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5		(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6		(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

2. Event C = "sum is 7." In the table above, these are the six cells on the diagonal from top-right to bottom-left: **(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)** — **6** outcomes.

(Same layout with **sums** in each cell: row + column = 7 exactly in those six cells.)

Die 1	Die 2	1	2	3	4	5	6
1		2	3	4	5	6	**7**
2		3	4	5	6	**7**	8
3		4	5	6	**7**	8	9
4		5	6	**7**	8	9	10
5		6	**7**	8	9	10	11
6		**7**	8	9	10	11	12

3. $P(C) = 6/36 = 1/6$.

Example: Lunch box mix-up

Suppose Allen, Beth and Cal are in an elevator, each holding a green lunchbox when an earthquake occurs. In the resulting chaos, their lunch boxes are dropped and shuffled around. Assuming they each pick up one lunch box, find the probability that they each pick up their own lunch box.

> **i** Worked solution — lunch boxes
>
> Label people A, B, C. List who gets which box as (Allen's, Beth's, Cal's): (A,B,C), (A,C,B), (B,A,C), (B,C,A), (C,A,B), (C,B,A) — **6** equally likely outcomes. Only (A,B,C) is favorable. $P = 1/6$.

Practice Problem: Three Coin Tosses

Suppose a fair coin is tossed three times. Let A be the event "the coin lands heads at least once". Find $P(A)$.

> **i** Worked solution — three coin tosses
>
> List S: **8** outcomes. Only TTT has no head $\rightarrow$ $P(\text{at least one head}) = 7/8$.

> Check yourself — Sample spaces and equally likely outcomes
>
> 1. What is $|S|$ when a fair die is rolled twice and outcomes are listed as ordered pairs?
> 2. State the probability rule when all outcomes in S are equally likely.
> 3. Two fair coin tosses: list S and find $P(\text{at least one tail})$.
> 4. Two dice: how many outcomes give a sum of 11? What is $P(\text{sum is 11})$?
> 5. Translate $P(A) = 0.7$ into a complete English sentence.
> 6. Three coin tosses: find $P(\text{at least one head})$.

Answers — Sample spaces and equally likely outcomes

1. List S as ordered pairs (i, j) with $i, j \in \{1, \dots, 6\}$ — **36** equally likely outcomes (list or count 6×6).
2. $P(A) = |A|/|S|$ — the number of outcomes in A divided by the number in S.
3. $S = \{\text{HH, HT, TH, TT}\}$. Event has 3 outcomes; $P = 3/4$.
4. List outcomes with sum 11: (5,6) and (6,5) — **2** out of 36. $P = 2/36 = 1/18$.
5. There is a 70% probability (or 0.7 chance) that event A occurs.
6. List S: **8** outcomes. Only TTT has no head $\to$ $P(\text{at least one H}) = 7/8$ (or $1 - 1/8$ by complement).

Section 3: Addition rule

Learning Outcomes:

1. State the **Addition Rule for any two events**
2. Use the addition rule to compute the probability of events A or B, being aware of special cases.

! Definition: Addition rule

For any two events A and B,

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Equivalently, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Visualizing the Addition Rule with a Venn Diagram

i Venn diagram idea

Draw overlapping circles for A and B. $P(A \text{ or } B)$ covers **both** circles but **subtract** the overlap $P(A \text{ and } B)$ so it is not counted twice.

Example: Suppose a class consists of 20 students. Of these students, 12 have backpacks and 3 have hats. There is only 1 student with both a backpack and a hat.

Suppose one student is randomly selected from the 20. What is the probability this student will either have a backpack OR have a hat?

i Worked solution — backpacks and hats (20 students)

$P(\text{backpack or hat}) = 12/20 + 3/20 - 1/20 = 14/20 = 0.7$.

Practice Problem: Suppose a class consists of 30 students. Of these students, 8 like to go sky diving and 20 like to go surfing. There are 7 students that like both sky diving and surfing.

Suppose one student is randomly selected from the 30. What is the probability this student will like sky diving OR surfing

> **Worked solution — sky diving or surfing**
>
> $P = 8/30 + 20/30 - 7/30 = 21/30 = 0.7$.

Special Cases of the Addition Rule

> **Definition: Complement rule**
>
> For any event A,
>
> $$P(\text{not } A) = P(A^c) = 1 - P(A)$$

> **Definition: Disjoint (non-overlapping) events**
>
> If A and B are **disjoint** (non-overlapping), then $P(A \text{ and } B) = 0$ and
>
> $$P(A \text{ or } B) = P(A) + P(B)$$

Example: Suppose the probability of it snowing at least once this November is 0.83. what is the probability that it will not snow at all this November?

> **Worked solution — complement**
>
> $P(\text{no snow}) = 1 - 0.83 = 0.17$.

Example: Suppose that a brand of gummy vitamins come in three animal shapes: Lions, tigers or bears . The company creates the gummies as follows: 30% of the shapes are bears, 30% are tigers and 40% are lions.

What is the probability that a randomly selected gummy will be either a bear or a tiger?

> **Worked solution — disjoint gummies**
>
> Bear and tiger are disjoint: $P = 0.30 + 0.30 = 0.60$.

Practice Problem: Suppose a card is drawn from a standard 52 card deck. What is the probability the card is...

a. A “face” card (king, queen or jack)?
b. An Ace?
c. Not an Ace?

Worked solution — card from deck

a. Face cards: $12/52 = 3/13$. b. Ace: $4/52 = 1/13$. c. Not an Ace: $48/52 = 12/13$.

Check yourself — Addition rule

1. State the **addition rule** for any events A and B.
2. State the **complement rule** for event A.
3. If A and B are disjoint, what is $P(A \text{ and } B)$? Write the simplified addition rule.
4. Class: 40 with backpacks, 10 with hats, 5 with both (out of 65). Find $P(\text{backpack or hat})$.
5. If $P(\text{rains at least once in July}) = 0.91$, find $P(\text{no rain in July})$.

Answers — Addition rule

1. $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.
2. $P(A^c) = P(\text{not } A) = 1 - P(A)$.
3. $P(A \text{ and } B) = 0$ and $P(A \text{ or } B) = P(A) + P(B)$.
4. $P(A) = 40/65$, $P(B) = 10/65$, $P(A \text{ and } B) = 5/65$. $P(A \text{ or } B) = 45/65 \approx 0.69$.
5. $P(\text{no rain}) = 1 - 0.91 = 0.09$.

Section 4: Conditional probability and independence

Learning Outcomes:

1. State the definition of **conditional probability** and compute $P(A \mid B)$.
2. Use the **multiplication rule** to compute $P(A \text{ and } B)$.
3. Determine whether two events are **independent** (using equivalent criteria).

> **!** Definition: Conditional probability
>
> **Notation:** $P(A \mid B)$ means "the probability event A occurs **given** event B occurred."
> **Definition:** For events A and B with $P(B) > 0$,
>
> $$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

Example 1 (die roll): Consider the trial of rolling a fair six-sided die once.

Let A be the event "the roll is a 3" and let B be the event "the roll is odd."

a. List the outcomes in A, in B, and in A and B.
b. Compute $P(A)$, $P(B)$, and $P(A \text{ and } B)$.
c. Find $P(A \mid B)$: the probability of getting a 3 **given** the roll resulted in an odd number.
d. Find $P(B \mid A)$: the probability of getting an odd number **given** the roll resulted in a 3.

> **i** Worked solution — die Example 1
>
> a. $A = \{3\}$, $B = \{1, 3, 5\}$, $A \cap B = \{3\}$ — 3 is the only roll in both events.
> b. $P(A) = 1/6$, $P(B) = 3/6 = 1/2$, and $P(A \text{ and } B) = 1/6$ (only outcome 3 satisfies both).
> c. **Conditional probability:** given the roll is odd, the sample space shrinks to $\{1, 3, 5\}$; one of those three is 3, so $P(A|B) = (1/6)/(1/2) = 1/3$. Notice $1/3 \neq P(A) = 1/6$ — knowing the roll is odd **changes** the chance of a 3, so A and B are **not independent**.
> d. $P(B|A) = 1$ because if you already know the roll is 3, it is odd with certainty.

Practice Problem 1: A jar contains 10 candies: 6 are red and 4 are blue. Two candies are selected **without replacement**.

Let A be the event "the first candy is red" and B be the event "the second candy is red."

a. Find $P(A)$.
b. Find $P(B \mid A)$.

c. Use the multiplication rule (below) to find $P(A \text{ and } B)$.

i Worked solution — candy jar

a. $P(A) = 6/10$ on the first draw.
b. **Given** the first candy is red, only 5 reds remain among 9 candies, so $P(B|A) = 5/9$. This is **not** the same as $P(B) = 6/10$ on a fresh first draw — the draws are **dependent** because you do not replace the candy.
c. Multiplication rule: $P(A \text{ and } B) = P(B|A)P(A) = (5/9)(6/10) = 1/3$.

! Definition: Multiplication rule

For any two events A and B with $P(B) > 0$,

$$P(A \text{ and } B) = P(A \mid B)\, P(B)$$

Equivalently (if $P(A) > 0$),

$$P(A \text{ and } B) = P(B \mid A)\, P(A)$$

Example 2 (two cards, no replacement): A card is drawn from a standard 52-card deck, then a second card is drawn **without replacement**.
Find the probability the **first** card is an Ace and the **second** card is a King.

Worked solution — Ace then King

First draw: $P(\text{Ace}) = 4/52$. **Given** an Ace was removed, 51 cards remain with 4 Kings, so $P(\text{King} \mid \text{Ace first}) = 4/51$. Multiply: $P = (4/52)(4/51) = 16/2652 = 4/663$. Without replacement, the second probability depends on the first — **dependent** trials.

Events A and B are **independent** if knowing that one event occurred does not change the probability of the other.
Equivalent ways to check independence:

- $P(A \mid B) = P(A)$ (when $P(B) > 0$)
- $P(B \mid A) = P(B)$ (when $P(A) > 0$)
- $P(A \text{ and } B) = P(A)\, P(B)$

Example 3 (king vs spade): Suppose a card is randomly dealt from a standard 52-card deck.
Is the event “the card is a king” independent of the event “the card is a spade”?

Worked solution — king vs spade

Let K = “the card is a king” and S = “the card is a spade.” One card is dealt from a standard 52-card deck.

Event	Count in deck	Probability
K	4 kings	$P(K) = 4/52 = 1/13$
S	13 spades	$P(S) = 13/52 = 1/4$
K and S	1 king of spades	$P(K \text{ and } S) = 1/52$

Check independence using $P(K \text{ and } S) = P(K)\,P(S)$:

$$P(K)\,P(S) = \frac{4}{52} \cdot \frac{13}{52} = \frac{52}{2704} = \frac{1}{52} = P(K \text{ and } S).$$

So **yes, K and S are independent** for a single random deal.

Equivalently: among spades, exactly one of 13 is a king, so $P(K \mid S) = 1/13 = P(K)$. Among kings, exactly one of 4 is a spade, so $P(S \mid K) = 1/4 = P(S)$.

Why independence matters for the binomial test

Independence is the **fourth BRP condition**. It means **learning the outcome of one trial does not change the probability on the next trial**.

- **Not independent:** drawing candies **without replacement** — if the first candy is red, fewer reds remain, so $P(B \mid A) \neq P(B)$.
- **Independent:** fair **coin tosses** — the coin has no memory; $P(H \text{ on toss } 2 \mid H \text{ on toss } 1) = P(H) = 1/2$.

Before you run a binomial test, you check all four BRP conditions. If trials are not independent (or if p changes from trial to trial), the null bar chart from `dbinom` describes the **wrong** imagined world for your data.

Tool for independence: compare $P(A \mid B)$ to $P(A)$, or check whether $P(A \text{ and } B) = P(A)P(B)$. If knowing that B occurred **changes** the probability of A, the events are **dependent** and a plain binomial model may not apply.

Multiplication Rule for Two Independent Events
If A and B are independent, then

$$P(A \text{ and } B) = P(A)\,P(B)$$

Example 4 (two coin tosses): Suppose a fair coin is tossed twice.

Find the probability of getting "heads" on the first toss and "tails" on the second toss.

Worked solution — two coin tosses (HT)

Let H_1 = heads on the first toss and T_2 = tails on the second toss. We want $P(H_1 \text{ and } T_2)$.
List the sample space with **rows = toss 1** and **columns = toss 2** (4 equally likely outcomes):

Toss 1	Toss 2	H	T
H		HH	**HT**
T		TH	TT

Only **HT** satisfies both events, so $P(H_1 \text{ and } T_2) = 1/4$.
The two tosses are **independent**, so you can also multiply: $P(H_1)\,P(T_2) = (1/2)(1/2) = 1/4$.

Practice Problem 2: Suppose a fair six-sided die is rolled twice.

Let A be the event "the first roll is even" and let B be the event "the second roll is 6."

a. Are A and B independent? (Explain briefly.)
b. Compute $P(A \text{ and } B)$.

Worked solution — two die rolls

a. **Yes, independent** — separate rolls do not affect each other; $P(B \mid A) = P(B)$.
b. $P(A \text{ and } B) = P(A)P(B) = (3/6)(1/6) = 1/12$ (even on roll 1 **and** 6 on roll 2).

Check yourself — Conditional probability and independence

1. In words, what does $P(A \mid B)$ mean?
2. Write the formula for $P(A \mid B)$ when $P(B) > 0$.
3. State the multiplication rule for $P(A \text{ and } B)$.
4. Give two equivalent ways to check that A and B are independent.
5. Fair die: find $P(\text{roll is } 2 \mid \text{roll is even})$.
6. Two fair coin tosses: find $P(\text{H on first and T on second})$.

Answers — Conditional probability and independence

1. The probability that A occurs **given that** B has already occurred.
2. $P(A \mid B) = P(A \text{ and } B)/P(B)$.
3. $P(A \text{ and } B) = P(A \mid B)\, P(B)$ (equivalently $P(B \mid A)\, P(A)$ when $P(A) > 0$).
4. $P(A \mid B) = P(A)$, or $P(A \text{ and } B) = P(A)\, P(B)$.
5. Given even $\{2, 4, 6\}$, one of three is 2: $P = 1/3$.
6. $P = (1/2)(1/2) = 1/4$ (independent trials).

Section 5: Counting and combinations

Learning Outcomes:

1. Distinguish between situations where **order matters** (permutations) and where **order does not matter** (combinations).
2. Use the formulas for ${}_nP_r$ and ${}_nC_r$ to solve counting problems.

Motivating Example: Ice Cream

Problem (a). Find the number of ways to distribute 3 distinct flavors of ice cream to 3 distinct bowls.

Worked solution — ice cream (a)

Label the flavors **C**hocolate, **V**anilla, and **S**trawberry. The three **bowls are distinct** (Bowl 1, Bowl 2, Bowl 3), so putting chocolate in Bowl 1 is different from putting chocolate in Bowl 2.

Multiply choices at each stage

1. **Bowl 1:** any of the 3 flavors → **3** choices.
2. **Bowl 2:** one flavor is already used → **2** choices left.
3. **Bowl 3:** only the remaining flavor fits → **1** choice.

$$\text{total ways} = 3 \times 2 \times 1 = 6.$$

Tree (each path = one full assignment)

```
Start
  Bowl 1: C   Bowl 2: V   Bowl 3: S   →  (C, V, S)
              Bowl 2: S   Bowl 3: V   →  (C, S, V)
  Bowl 1: V   Bowl 2: C   Bowl 3: S   →  (V, C, S)
              Bowl 2: S   Bowl 3: C   →  (V, S, C)
  Bowl 1: S   Bowl 2: C   Bowl 3: V   →  (S, C, V)
              Bowl 2: V   Bowl 3: C   →  (S, V, C)
```

Six branches → **6** ways. The tree shows the same idea as $3 \times 2 \times 1$: at the first split you have 3 options, then 2, then 1.

List all 6 assignments

Bowl 1	Bowl 2	Bowl 3
C	V	S
C	S	V
V	C	S

V	S	C
S	C	V
S	V	C

What is a factorial?
When you place n **distinct objects into** n **distinct slots** and use each object once, you multiply the number of choices at each stage:

$$n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1.$$

That product is written $n!$ (read "n factorial"). Here $n = 3$, so

$$3! = 3 \times 2 \times 1 = 6.$$

Convention: $0! = 1$ (defined so counting formulas still work when nothing is left to arrange). The formal definition and more examples are in **Factorial Notation** below.

Problem (b). Find the number of ways to choose 2 distinct flavors of ice cream from 26 flavors to create an ice cream cone with one scoop on the top and one scoop on the bottom.

ℹ Worked solution — ice cream (b)

Top and bottom are different positions, so chocolate on top / vanilla on bottom is **not** the same cone as vanilla on top / chocolate on bottom. **Order matters** → count **permutations**.

Multiply choices at each stage (26 flavors)

1. **Top scoop:** any of 26 flavors → **26** choices.
2. **Bottom scoop:** must be a **different** flavor → **25** choices left.

$$\text{total cones} = 26 \times 25 = 650 = {}_{26}P_2.$$

Full tree with 3 flavors (illustration)

The problem has 26 flavors; with **C**, **V**, **S** only, every branch is listed below. The same pattern gives 26×25 when there are 26 flavors.

```
Start
  Top: C   Bottom: V  →  (C on top, V on bottom)
             Bottom: S  →  (C on top, S on bottom)
  Top: V   Bottom: C  →  (V on top, C on bottom)
             Bottom: S  →  (V on top, S on bottom)
  Top: S   Bottom: C  →  (S on top, C on bottom)
             Bottom: V  →  (S on top, V on bottom)
```

6 branches → $3 \times 2 = 6 = {}_3P_2$. Each ordered pair appears once.

List all 6 ordered cones (3 flavors)

Top	Bottom
C	V
C	S
V	C
V	S

S	C
S	V

With **26** flavors: $26 \times 25 = 650$ cones (too many to list, but the tree has 26 branches at the first split and 25 at each second split).

Problem (c). Find the number of ways to choose 2 distinct flavors of ice cream from 26 flavors to create a dish with two scoops of ice cream, where the order of the scoops doesn't matter.

ℹ Worked solution — ice cream (c)

A dish is just a set of two flavors — chocolate + vanilla is the same dish as vanilla + chocolate. **Order does not matter** → count **combinations**.

Step 1: List ordered scoops (same tree as part b)

If we temporarily treat "scoop 1" and "scoop 2" as labeled positions, we get the same **6** ordered outcomes as in part (b):

Scoop 1	Scoop 2
C	V
C	S
V	C
V	S
S	C
S	V

Step 2: Group equivalent dishes (swap the two scoops)

Each **unordered pair** appears **twice** in the table — once in each order:

Dish (flavors only)	Ordered versions (grouped)	Count in group
{C, V}	(C, V) and (V, C)	2
{C, S}	(C, S) and (S, C)	2
{S, V}	(S, V) and (V, S)	2

3 distinct dishes → $\frac{6}{2} = 3 = {}_3C_2$.

```
From part (b) tree - merge swapped branches:
  (C,V)  ↔  (V,C)   } one dish {C, V}
  (C,S)  ↔  (S,C)   } one dish {C, S}
  (S,V)  ↔  (V,S)   } one dish {S, V}
```

Step 3: Scale to 26 flavors

Part (b) gave **650** ordered cones (${}_{26}P_2$). Each unordered pair of flavors is counted **twice** (flavor A on top / B on bottom, and B on top / A on bottom), so

$$\text{dishes} = \frac{650}{2} = 325 = \frac{26 \times 25}{2} = {}_{26}C_2.$$

Equivalently: ${}_{26}C_2 = \frac{{}_{26}P_2}{2!} = \frac{650}{2} = 325.$

Key Idea: Does Order Matter?

! Definitions: Permutation and combination

- **Permutation:** An **arrangement** of objects where **order matters**. (Chocolate on top, vanilla on bottom $\neq$ vanilla on top, chocolate on bottom.)
- **Combination:** A **selection** of objects where **order does not matter**. (Chocolate + vanilla in a dish = vanilla + chocolate.)

! Definition: Factorial

The **factorial** of a nonnegative integer n, written $n!$, is the product of all positive integers from n down to 1:

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$$

Convention: $0! = 1$. (This is defined so that our permutation and combination formulas work correctly when $r = n$ or $r = 0$.)

Examples:

- $3! = 3 \times 2 \times 1 = 6$
- $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
- $1! = 1$

! Formulas: Permutations and combinations

Permutations ${}_nP_r$ (or $P(n,r)$): The number of ways to arrange r objects chosen from n distinct objects, where **order matters**.

$${}_nP_r = \frac{n!}{(n-r)!} = n(n-1)(n-2)\cdots(n-r+1)$$

Combinations ${}_nC_r$ (or $\binom{n}{r}$ or $C(n,r)$): The number of ways to choose r objects from n distinct objects, where **order does not matter**.

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{{}_nP_r}{r!}$$

Relationship: ${}_nC_r = \frac{{}_nP_r}{r!}$ — we divide by $r!$ because each combination of r objects can be arranged in $r!$ different orders, and we want to count them once.

Second Example: Choosing 3 Letters from {A, B, C, D, E}

Suppose we have 5 letters: A, B, C, D, E. We want to choose 3 of them.

Part 1 (order matters): How many ways can we choose 3 letters if the order of our choice matters? (e.g., for a 3-letter code)?

> **Worked solution — letters, order matters**
>
> ${}_5P_3 = 5 \times 4 \times 3 = 60.$

Part 2 (order doesn't matter): How many ways can we choose 3 letters if the order of our choice does not matter (e.g., for a committee of 3)?

> **Worked solution — letters, committee**
>
> ${}_5C_3 = 10.$

Summary

Rule of thumb: Ask yourself: "If I swap two of the chosen items, do I get a different outcome?" If yes → permutation. If no → combination.

> Check yourself — Counting and combinations
>
> 1. What is the difference between a **permutation** and a **combination**?
> 2. Compute 4! and state the value of 0!.
> 3. Compute ${}_5C_2$ and ${}_5P_2$.
> 4. How many ways to choose 3 people from 7 for a committee (order does not matter)?
> 5. Why does ${}_nC_r =_n P_r/r!$?

Answers — Counting and combinations

1. **Permutation:** order matters. **Combination:** order does not matter (a selection).
2. $4! = 24$; $0! = 1$ by convention.
3. ${}_5C_2 = 10$; ${}_5P_2 = 20$.
4. $\binom{7}{3} = 35$.
5. Each combination of r objects can be arranged in $r!$ orders; dividing counts each selection once.

Section 6: Random variables and the binomial process

Learning Outcomes:

1. Define **random variable** and its **probability distribution function (pdf)** (for discrete random variables).
2. Define a **binomial random variable** and a **binomial random process**.
3. Match the notation (n, x, p) and **four conditions** for a binomial process with the process of tossing a coin.
4. Express probabilities for binomial random variables via the binomial pdf formula.

Random variable

A **random variable** is defined as…

Probability distribution function (pdf)

> **!** Definition: Probability distribution function (pdf)
>
> For a discrete random variable X, the **probability distribution function** (or **probability mass function**) gives the probability that X takes each possible value. We write $f(x) = P(X = x)$ for each possible value x.

Examples: Coin tosses

Example 1. A coin is tossed $n = 1$ time; $p =$ probability of heads. Find an expression for $f(x) = P(X = x)$ for $x = 0$ and $x = 1$.

ℹ Worked solution — $n = 1$

Random variable: Let X = number of heads in 1 toss. Possible values: $x = 0$ or $x = 1$.
Step 1 — List the sample space S

Outcome	X
H (heads)	1
T (tails)	0

$|S| = 2$ equally likely *types* of outcome if the coin is fair; in general $P(\text{H}) = p$ and $P(\text{T}) = 1 - p$.
Step 2 — Probability of each outcome

- $P(\text{H}) = p$
- $P(\text{T}) = 1 - p$

(No multiplication rule needed yet — only one toss.)
Step 3 — Build $f(x) = P(X = x)$

x	Outcomes with $X = x$	$f(x)$
0	T	$f(0) = 1 - p$
1	H	$f(1) = p$

Check: $f(0) + f(1) = (1 - p) + p = 1$.
Step 4 — One expression using exponents (the "exponent trick")
Here $n = 1$, so each outcome has either 0 or 1 head. For a given x:

- multiply p x **times** (once per head) $\to p^x$
- multiply $(1 - p)$ $(1 - x)$ **times** (once per tail) $\to (1 - p)^{1-x}$

x	Heads / tails on the one toss	Exponent form	Matches Step 3?
0	0 H, 1 T	$p^0(1 - p)^1$	$f(0) = 1 - p$
1	1 H, 0 T	$p^1(1 - p)^0$	$f(1) = p$

Convention: $p^0 = 1$ and $(1 - p)^0 = 1$ (so a zero exponent does not zero out the probability).
There is only **one** outcome for each x, so the pdf is exactly that product:

$$f(x) = p^x(1 - p)^{1-x}, \quad x = 0 \text{ or } 1.$$

Example 2. A coin is tossed $n = 2$ times; p = probability of heads. Find an expression for $f(x)$ for $x = 0, 1, 2$.

ℹ Worked solution — $n = 2$

Random variable: X = number of heads in 2 tosses; $x \in \{0, 1, 2\}$.
Step 1 — List S (rows = toss 1, columns = toss 2)

Toss 1	Toss 2	H	T
H		HH	HT
T		TH	TT

All **4** ordered outcomes: HH, HT, TH, TT.

Step 2 — Multiply along each sequence (independence)

Outcome	Calculation	Probability
HH	$P(\mathrm{H}) \cdot P(\mathrm{H})$	p^2
HT	$p(1-p)$	$p(1-p)$
TH	$(1-p)p$	$p(1-p)$
TT	$(1-p)^2$	$(1-p)^2$

Step 3 — Exponent trick ($n = 2$)

Example 1 used exponents x and $(1-x)$ for one toss. With **two** tosses, a sequence with x heads and $(2-x)$ tails has

$$p^x(1-p)^{2-x}.$$

x	Example sequence	As $p^x(1-p)^{2-x}$	Matches Step 2?
0	TT	$p^0(1-p)^2$	$(1-p)^2$
1	HT	$p^1(1-p)^1$	$p(1-p)$
2	HH	$p^2(1-p)^0$	p^2

Step 4 — Group by x (addition rule)

x	Outcomes	Count	$f(x) = P(X = x)$
0	TT	1	$(1-p)^2$
1	HT, TH	**2**	$p(1-p) + p(1-p) = 2p(1-p)$
2	HH	1	p^2

Step 5 — Count with combinations, then one formula for $f(x)$

Several sequences can share the same x; each has probability $p^x(1-p)^{2-x}$. Count them with $\binom{2}{x}$:

$$f(x) = \binom{2}{x} p^x(1-p)^{2-x}, \quad x = 0, 1, 2.$$

For example, $f(1) = \binom{2}{1}p^1(1-p)^1 = 2p(1-p)$ from outcomes HT and TH.

Example 3. A coin is tossed $n = 3$ times; p = probability of heads. Find an expression for $f(x)$ for $x = 0, 1, 2, 3$.

ℹ Worked solution — $n = 3$

Random variable: X = number of heads in 3 tosses; $x \in \{0, 1, 2, 3\}$.

Step 1 — Tree / list all of S ($|S| = 8$)

```
Toss 1: H   Toss 2: H   Toss 3: H  →  HHH
                 Toss 3: T  →  HHT
           Toss 2: T   Toss 3: H  →  HTH
                 Toss 3: T  →  HTT
Toss 1: T   Toss 2: H   Toss 3: H  →  THH
                 Toss 3: T  →  THT
           Toss 2: T   Toss 3: H  →  TTH
                 Toss 3: T  →  TTT
```

Step 2 — Exponent trick ($n = 3$)
Building on the exponents from earlier examples:

n	Tail exponent on $(1-p)$	One sequence with x heads
1 (Ex. 1)	$1-x$	$p^x(1-p)^{1-x}$
2 (Ex. 2)	$2-x$	$p^x(1-p)^{2-x}$
3 (here)	$3-x$	$p^x(1-p)^{3-x}$

So any **one** of the 8 sequences with x heads has probability $p^x(1-p)^{3-x}$.
Step 3 — Group by x

x	Outcomes (all listed)	How many?	$f(x)$
0	TTT	1	$(1-p)^3$
1	HTT, THT, TTH	**3**	$3\,p(1-p)^2$
2	HHT, HTH, THH	**3**	$3\,p^2(1-p)$
3	HHH	1	p^3

Step 4 — Combinations count the groups
Choose which x of the 3 tosses are heads: $\binom{3}{x}$ ways. Multiply by the probability of **one** such sequence:

$$f(x) = \binom{3}{x} p^x(1-p)^{3-x}, \quad x = 0, 1, 2, 3.$$

Explicitly: $f(0) = \binom{3}{0}(1-p)^3$, $f(1) = \binom{3}{1}p(1-p)^2$, $f(2) = \binom{3}{2}p^2(1-p)$, $f(3) = \binom{3}{3}p^3$.

Example 4. A coin is tossed $n = 4$ times; p = probability of heads. Find an expression for $f(x)$ for $x = 0, 1, 2, 3, 4$.

ℹ Worked solution — $n = 4$

Random variable: X = number of heads in 4 tosses; $x \in \{0, 1, 2, 3, 4\}$.
Step 1 — Size of S
Four independent tosses → $2 \times 2 \times 2 \times 2 = 2^4 = 16$ ordered outcomes (too many to write in a table, but we can count by structure).
Step 2 — Exponent trick ($n = 4$)
For $n = 4$ tosses, a sequence with x heads and $(4-x)$ tails has

$$P(\text{one such sequence}) = p^x(1-p)^{4-x}.$$

(Compare $1-x$, $2-x$, $3-x$ in Examples 1–3.)
Step 3 — How many sequences for each x? (combinations)

Pick **which** x **of the 4 tosses** are heads; order within heads/tails is already fixed once those positions are chosen.

x	$\binom{4}{x}$ = # sequences	One example sequence	$f(x) = \binom{4}{x} p^x (1-p)^{4-x}$
0	$\binom{4}{0} = 1$	TTTT	$(1-p)^4$
1	$\binom{4}{1} = 4$	HTTT, THTT, TTHT, TTTH	$4\,p(1-p)^3$
2	$\binom{4}{2} = 6$	HHTT, HTHT, HTTH, THHT, THTH, TTHH	$6\,p^2(1-p)^2$
3	$\binom{4}{3} = 4$	HHHT, HHTH, HTHH, THHH	$4\,p^3(1-p)$
4	$\binom{4}{4} = 1$	HHHH	p^4

Step 4 — Single formula for this coin experiment

$$f(x) = P(X = x) = \binom{4}{x} p^x (1-p)^{4-x}, \quad x = 0, 1, 2, 3, 4.$$

The coefficients **1, 4, 6, 4, 1** are exactly $\binom{4}{x}$ — they count how many ways to place x heads among 4 tosses, each contributing $p^x(1-p)^{4-x}$.

General case. A coin is tossed n times; p = probability of heads. Find an expression for $f(x) = P(X = x)$ for $x = 0, 1, 2, \ldots, n$.

Worked solution — general n

Examples 1–4 built the tail exponent toss by toss:

n	One sequence with x heads
1	$p^x(1-p)^{1-x}$
2	$p^x(1-p)^{2-x}$
3	$p^x(1-p)^{3-x}$
4	$p^x(1-p)^{4-x}$

For **any** fixed n: x heads and $(n-x)$ tails on independent tosses give

$$P(\text{one sequence}) = p^x(1-p)^{n-x}.$$

There are $\binom{n}{x}$ different sequences with exactly x heads, so

$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \ldots, n.$$

Binomial random variable and binomial random process

A **binomial random variable** X counts the number of "successes" in a **binomial random process**.

Definition: Four BRP conditions

Four conditions for a binomial random process:

1. **Fixed number of trials** n (e.g., n coin tosses).
2. **Two outcomes** on each trial: "success" or "failure" (e.g., heads vs tails).
3. **Independent trials** — the outcome of one trial does not affect the others.
4. **Constant probability** p of success on each trial (e.g., same p for every toss).

Notation: n = number of trials, p = probability of success on one trial, X = number of successes, and x = a specific value of X (0, 1, 2, ..., n).

Binomial formula

Definition: Binomial pmf

If X is binomial with n trials and success probability p, then for $x = 0, 1, 2, \ldots, n$,

$$P(X = x) = f(x) = \binom{n}{x} p^x (1-p)^{n-x} =_n C_x \, p^x (1-p)^{n-x}$$

where $\binom{n}{x} =_n C_x = \dfrac{n!}{x!(n-x)!}$ is the number of ways to choose x successes out of n trials.

---##

Check yourself — Random variables and the binomial process

1. What is a **random variable**?
2. For discrete X, what does the pmf $f(x) = P(X = x)$ mean?
3. List the **four BRP conditions**.
4. In words, what do n, p, X, and x mean in a binomial setting?

Answers — Random variables and the binomial process

1. A rule that assigns a **number** to each outcome in a sample space (written with capitals such as X).
2. $f(x)$ is the probability that the random variable X equals the specific value x.
3. (a) Binary outcome each trial. (b) Fixed n. (c) Fixed p each trial. (d) Independent trials.
4. n = number of trials; p = success probability on one trial; X = random count of successes; x = a specific possible value of X.

Section 7: Binomial pmf

❗ Definitions and formula summary (binomial)

Notation:

- n = number of trials
- p = probability of "success" on a single trial
- x = observed number of successes in n trials
- X = random variable that counts the number of successes in n trials

Factorial: $n! = n(n-1)(n-2)\cdots(3)(2)(1)$

Counting:

- ${}_nC_x = \binom{n}{x} = \dfrac{n!}{x!(n-x)!}$
- ${}_nP_x = \dfrac{n!}{(n-x)!}$

Four conditions for a binomial random process (BRP):

1. Each trial results in either "success" or "failure".
2. Outcomes between trials are **independent**.
3. The probability of success p is the same on every trial.
4. A **fixed** number of trials n are observed.

Binomial random variable: X = number of successes in n trials when data are collected from a BRP.
Probability distribution function (pdf) for a binomial random variable X:

$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x} =_n C_x\, p^x (1-p)^{n-x}$$

Cumulative probability: $P(X \le x) = f(0) + f(1) + f(2) + \cdots + f(x)$
Note: $P(X > x) = 1 - P(X \le x) = f(x+1) + f(x+2) + \cdots + f(n)$
Mean of X: $\mu = np$
Standard deviation of X: $\sigma = \sqrt{np(1-p)}$

Practice problems

1. Which situation is binomial?

Which of the following situations is an example of data being collected from a **binomial random process**? Be prepared to justify your answer by discussing the validity of each of the four conditions.

a. Rolling a six-sided die 15 times and counting the number of times a "3" was obtained.
b. Picking 10 gummy vitamins from a bottle that contains 100 vitamins of various animal shapes. Count the number

of lions, tigers, and bears in the set of 10.

c. Drawing cards one at a time from a deck of 52, counting the number of kings drawn. After each card is drawn, it is **replaced** in the deck, the deck is shuffled, and another card is drawn.
d. Drawing cards one at a time from a deck of 52, counting the number of kings drawn. After each card is drawn, it is **not** replaced in the deck.
e. Giving a class activity to 20 students and counting the number of these students who ask a question about it during class.

> **i** Worked solution — which is binomial?
>
> **(c) is binomial** (fixed n, binary king/not, replacement keeps p constant, independent draws). (a) can be binomial if success = "3". (b) is **not** (more than two categories). (d) is **not** (p changes). (e) is **not** a standard BRP (trials not identical random process).

---

2. Identify n, p, and X; write probability expression

For each situation below, identify n, p, and the appropriate value to compare to X, then write an expression for the probability using the pdf of a binomial random variable.

Problem 2. Suppose Bob flips a fair coin 4 times. What is the probability he gets exactly 2 heads?

> **i** Worked solution — exactly 2 heads
>
> $n = 4$, $p = 0.5$, $P(X = 2) = \binom{4}{2}(0.5)^4 = 6/16 = 3/8$.

Problem 3. Suppose Neal has an unfair coin: it has an 80% chance of landing "heads". What is the probability that he gets exactly 2 heads if he tosses the coin 4 times?

> **i** Worked solution — unfair coin
>
> $P(X = 2) = \binom{4}{2}(0.8)^2(0.2)^2 = 6(0.64)(0.04) = 0.1536$.

Problem 4. Suppose that in a week of 7 days, there is a 65% chance of rain on any single day. What is the probability of it raining on **at least** 2 of the 7 days? On **at most** 2 of the 7 days?

> **i** Worked solution — rain days
>
> $n = 7$, $p = 0.65$. **At least 2:** $P(X \geq 2) = 1 - P(X \leq 1)$. **At most 2:** $P(X \leq 2) = f(0) + f(1) + f(2)$ (or `pbinom(2,7,0.65)`).

Problem 5. Suppose customers at a coffee shop can order either iced or hot coffee. Sharon observes 6 customers ordering one day: of these, only 2 order iced coffee. If 10% of all customers order iced coffee, what is the probability of observing 2 or more iced orders (under this model)?

ℹ Worked solution — iced coffee

$n = 6$, $p = 0.10$. $P(X \geq 2) = 1 - P(X \leq 1)$ (or `1-pbinom(1,6,0.10)`).

Problem 6. Compute the mean and standard deviation of the random variable X that is the number of "heads" in 25 tosses of a fair coin.

ℹ Worked solution — mean and SD

$\mu = np = 25(0.5) = 12.5$. $\sigma = \sqrt{np(1-p)} = \sqrt{25(0.5)(0.5)} = 2.5$.

Check yourself — Binomial pmf

1. Write the **binomial pmf** $P(X = x)$.
2. A fair coin is flipped 12 times. Identify n, p, and what x represents.
3. 20% of seeds fail to germinate; you plant 8 seeds. Identify n and p if $X =$ number that fail.
4. Write $\binom{n}{x}$ in factorial form.

Answers — Binomial pmf

1. $P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$ for $x = 0, 1, \dots, n$.
2. $n = 12$ trials; $p = 0.5$; x is a possible number of heads (0 through 12).
3. $n = 8$, $p = 0.20$.
4. $\binom{n}{x} = n!/[x!(n - x)!]$.